

From Graphical Models to Generative AI

Mathematical Foundations, Algorithms, and Modern Applications

Michael (Misha) Chertkov

MATH 577-001 University of Arizona

Lecture 1 — One Mathematical Picture

Monday, August 24, 2026 Mathematics 514

Living-book companion: *From Graphical Models to Structured Generative AI*

Why structure?

A generic law on n binary variables

$$p(x_1, \dots, x_n)$$

needs up to $2^n - 1$ parameters.

Structured probability

$$p(x) = \frac{1}{Z} \prod_{a \in F} \psi_a(x_{\partial a})$$

exposes dependencies, constraints, and computation.

**Graphical models are not mainly about drawing graphs.
They are about exploiting structure in otherwise impossible
high-dimensional computations.**

One structured object — six computational questions

$$p(x) = \frac{1}{Z} \prod_a \psi_a(x_{\partial a})$$

1. Normalize	2. Infer	3. Optimize
$Z = \sum_x \prod_a \psi_a(x_{\partial a})$	$p_i(x_i), p_{ij}(x_i, x_j), \dots$	$x^* = \arg \max_x p(x)$
4. Sample	5. Learn	6. Generate
$x^{(1)}, x^{(2)}, \dots \sim p$	data $\longrightarrow$ factors / parameters / structure	construct a process producing new structured samples

Same probability object; different computational tasks.

A Rosetta Stone: many languages for structured probability

Framework	Structure	Computational object
Factor graph / MRF	local factors	messages, beliefs, Z
Tensor network	local tensors / bonds	partial contractions
Energy-based model	energy $E(x)$	Z , gradients, samples
Latent / VAE	latent-variable graph	posterior / ELBO
Diffusion / bridge	space-time path structure	score, reverse drift, control
Autoregressive model	ordered conditionals	$p(x_i \mid x_{<i})$

Graphical Models $\longrightarrow$
Neuralization

Generative AI $\longrightarrow$
Structuralization

Three mathematical spines

I. Factors $\rightarrow$ messages $\rightarrow$ gauges $\rightarrow$ loops / tensors

Static structured probability: exact and approximate computation, invariants, corrections.

II. Markov dynamics $\rightarrow$ bridges $\rightarrow$ control / generation

Dynamic structured probability: paths, conditioning, transport, reverse dynamics.

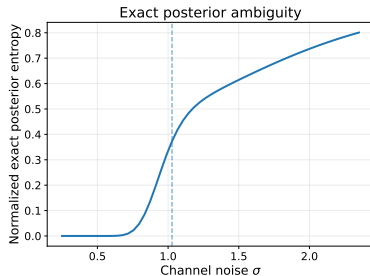
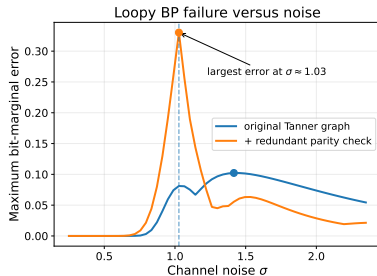
III. Structured reference + explicit correction

$$\frac{dP}{dQ_\theta} = \frac{W_\theta}{\mathbb{E}_{Q_\theta}[W_\theta]} \implies \text{choose } Q_\theta \text{ so that the residual is simple.}$$

**Approximate only what must be approximated; retain
correction/verification when possible.**

An approximation is not understood until we know how it fails

Push the method:



easy



crossover



failure / loss of
guarantee

What observable tells us that breakdown is coming?

How this course works: solve, verify, challenge, extend

No examinations

Exercise 1

Problem 2

Challenge 3

Presentation 4

1. Submit by the deadline.
2. Instructor screens; approved work is posted anonymously.
3. Others may challenge: error, missing assumption, counterexample, stronger result, better stress test.
4. Book derivations and notebooks are challengeable too.

AI tools are allowed for course work, but you own the mathematics: disclose substantial use, verify it, and be prepared to explain it.

Five research studios — individual projects, shared neighborhoods

1. **Quantum, tensors, and loop structure**
BP, gauges, loop calculus, tensor networks, quantum/classical structure.
2. **Neural and biological systems**
latent states, learned messages/operators, neural and biological dynamics.
3. **Energy and networked systems**
Gaussian/network inference, stochastic flows, state estimation, restoration/control.
4. **Continuous structured GenAI**
diffusion, bridges, Feynman–Kac, stochastic/path-integral control, PoE/PoIE/PoPE.
5. **Discrete GenAI and language**
coding, discrete diffusion, autoregressive generation, Sampling Decisions, LLM structure.

A studio is a research community — not a group project.

Before Wednesday: where is the structure in your problem?

Where is the structure in your problem?

1. What are the variables / states?
2. What interacts, constrains, or couples them?
3. What is uncertain or only partially observed?
4. What do you need to **infer, optimize, learn, sample, or generate**?

**Research-interest note due Friday, Aug 28, 6 PM.
On Wednesday, Aug 26, we start setting up graphical-model
language.**